TBDA Sprint 2 — Tasks 7 & 8

*Summary of the README files and guide to the delivery package **Task7&8_rev0_19-10** (CeDInt Storey 1, energy forecasting) · 19 October 2026*

*Points requiring additional discussion / information from the product owner were **highlighted yellow and bolded.***

At a glance

Task 7 (baselines):

- Two simple forecasts (e.g. Baselines)
  - B1. Same hour last week
  - B2. Weekday × hour median
  - Both scored on the validation and test week for hourly energy of the 8 eligible circuits
- => **B2, weekday × hour median, is the stronger reference**: lower error on 8 of 8 circuits in validation and 6 of 8 in test.

Task 8 (first ML model):

- Chosen model type: **gradient boosting**
- One shared model for all 8 circuits per horizon (+1h, +6h, day-ahead)
- Trained on the training period only, chosen on validation, frozen, then run once on the test week against the baselines
  - **New baseline, B3: *Persistence*:** Estimates values as same as in the last hour before forecast

Result:

- B1 & B2
  - ML model clearly beats the Task 7 baselines at +1h (7 of 8 circuits, 8 of 8 days in both weeks)
  - At +6h and day-ahead it matches but does not yet beat B2 (weekday × hour median)
- B3
  - Better on validation
  - Roughly level on the test week

Reproducibility:

- Full rebuild from the frozen file reproduces the dataset, model choice, predictions and results exactly
- The model cannot be re-tuned after freezing.

How the pieces connect

The frozen source file (Task 4) is turned into an hourly dataset (v1.1). The Task 7 baselines and the Task 8 model both read that dataset, use the same forecast timing and are scored on exactly the same hours, so their numbers are directly comparable.

Stage	Script	Input	Output
1 Hourly dataset	`run_task7.py`	frozen file (data/)	derived/ (dataset v1.1, eligibility)
2 Baselines (Task 7)	`run_task7_baselines.py`	derived/	results/task7_*
3 Train + choose + freeze	`run_task8_tune.py`	derived/ (train + validation only)	models/, results/task8_tuning_log, validation results
4 Test run (once)	`run_task8_test.py`	models/ + derived/	results/task8_results, test predictions, comparison
5 Error analysis	`run_task8_analysis.py`	results/, models/	results/task8_error_analysis, task8_tables.md

1 · Hourly dataset v1.1 (README_Task7.md)

The 2025 data are minute-level; forecasts are hourly. The pipeline builds one row per circuit and hour (16 circuits × 1,341 hours = 21,456 rows) without ever changing the frozen file (checked by SHA-256 before and after the run).

Step	What it does
S05 Load	Reads the frozen file, verifies its checksum, adds UTC and Madrid timestamps and the circuit ID (zone|sensor).
S06 Energy	Sums minute energy per hour. An hour is valid with at least 30 readings and no long gap. Invalid hours are left empty, never 0.
S07 Power, current	Time-weighted hourly means of W and I (readings are snapshots, so they are averaged, not summed).
S08 Split, calendar	Labels train / validation / test (Task 6 dates, UTC) and adds hour, weekday, month, weekend, holiday (Madrid time).
S09 Save	Writes the versioned dataset plus a META file with checksums and settings; refuses to overwrite.
S10 Eligibility	A circuit needs at least 90% valid hours in validation and test: 8 of 16 qualify (all NE and NW).
S11 Metrics	MAE and RMSE that skip missing values and report how many hours were scored.

Key findings

- Each energy value covers the whole time since the previous reading, so an hour stays complete even with missing rows (median 48 readings per hour, never 60).
- After an outage, one reading carries the whole outage (e.g. 10,229 Wh after 1,006 minutes on 17 Sep): that hour is invalid. **New in v1.1:** the hour *before* an outage is also invalid if more than 5 minutes at its end fall inside the gap. 96 hours change, all in training; eligibility is unchanged.
- SE and SW are excluded. Beyond missing hours, their energy and power readings disagree (e.g. SE|Lighting up to 2,097 Wh at 0.5 W), **needs to be discussed with the data owner.**
- No public holiday falls in the 2025 window; the UPM academic calendar is not yet used. No daylight-saving change in the window, so every week has 168 hours.

Settings still to confirm with the team

`ENERGY_MAX_GAP = 15` min, `ENERGY_TAIL_MAX = 5` min, `MAX_GAP = 5` min, `MIN_COVERAGE_MIN = 30` min; `THRESHOLD = 30` readings follows the Task 6 coverage analysis. All live in config.py; changing one means a new dataset version.

2 · Task 7 — Baseline forecasts (README_Task7_baselines.md)

Decisions taken on 18 Oct

- **Day-ahead forecasts start at midnight Madrid time** and cover that local day; split dates stay in UTC (Task 6). Every hour ahead (1–24) then has exactly 7 targets per week.
- **Leakage rule (protocol §9), reworded:** forecasts may use any value observed before the forecast is made, including validation and test values; validation and test data are never used to fit, tune or select anything.
- Both baselines are scored on the same hours (where both can forecast). Baseline 1 can forecast only 84% of validation hours, because the 17 Sep outage falls in the last training week.

Results — MAE in Wh per hour, day-ahead (better baseline in bold)

Circuit	Avg. Wh/h (val)	B1 val	B2 val	Avg. Wh/h (test)	B1 test	B2 test
NE|HVAC	2.7	0.72	**0.55**	12.5	10.44	**10.39**
NE|Plugs & Sockets	186.4	7.84	**6.62**	189.6	5.65	**3.93**
NW|Emerg. Lighting	10.9	1.04	**0.97**	39.6	**29.30**	29.72
NW|HVAC	33.7	137.54	**77.47**	70.7	**37.69**	43.86
NW|Hand Dryer	0.8	1.16	**1.00**	0.6	1.15	**0.77**
NW|Lighting	63.4	55.43	**35.58**	83.7	61.81	**49.02**
NW|Plugs & Sockets	805.9	324.76	**307.96**	763.3	355.93	**265.15**
NW|RACK	186.5	4.97	**3.73**	187.1	3.72	**3.30**

B1 = same hour last week; B2 = weekday × hour median from training data. Baseline errors are the same at every horizon, because neither uses information that arrives closer to the target hour.

Interpretation

- **B2 is the reference to beat**: better on 8 of 8 circuits in validation, 6 of 8 in test, and it always produces a forecast.
- Stable loads are easy (NW|RACK, NE|Plugs & Sockets: errors of 2–4% of average use); plugs and lighting are moderate (35–88%); the hand dryer is too small and irregular to say much.
- The largest errors come from changes in operation: NW|HVAC was on standby all validation week; NE|HVAC genuinely ran for about 3 hours on 3 Oct.
- NW|Emerg. Lighting has 18 test hours on 5–6 Oct with energy 10–45× the measured power. **A data error, kept in the official results and flagged for the data owner.**

3 · Task 8 — First ML model (README_Task8.md)

Setup

- **Model:** scikit-learn HistGradientBoostingRegressor, absolute-error loss (MAE is the primary metric), one model per horizon. One shared (pooled) model for all 8 circuits beat one model per circuit on validation.
- **Inputs**, all known when the forecast is made: last valid hour; mean and spread of the last 24 h; mean of the last 168 h; same hour 24 h and 168 h before the target; median of the same hour over 4 weeks; hour, weekday, weekend; circuit; hours ahead. Values are divided by each circuit's training average.
- **Fourth reference (B3, persistence):** the last valid hour before the forecast is made.
- **Protocol:** fitted on training data only, no refit on training + validation (§4/§8/§9). Settings chosen on validation with a rule fixed beforehand; the frozen model is exactly the validated one; the test week was run once.

Frozen configuration

Horizon	Model	Learning rate	Leaves	Min. leaf	Validation score	Training hours
+1h	pooled	0.05	15	50	0.254	5,681
+6h	pooled	0.05	15	20	0.360	5,641
day-ahead	pooled	0.1	15	50	0.384	5,665

Results — scaled MAE (MAE ÷ circuit training average, mean over 8 circuits; lower is better)

Week	Horizon	B1	B2	B3	ML	Best	ML better than B2	ML better than B3
validation	+1h	0.578	0.427	0.290	0.246	ML	7/8	4/8
validation	+6h	0.578	0.429	0.460	0.355	ML	7/8	5/8
validation	day-ahead	0.578	0.427	0.367	0.384	B3	6/8	5/8
test	+1h	1.266	1.166	0.930	1.032	B3	7/8	4/8
test	+6h	1.267	1.167	1.887	1.175	B2	4/8	8/8
test	day-ahead	1.267	1.166	1.162	1.173	B3	5/8	3/8
test without 2 events	+1h	0.490	0.389	0.336	0.265	ML	7/8	6/8
test without 2 events	+6h	0.491	0.390	0.698	0.398	B2	4/8	7/8
test without 2 events	day-ahead	0.491	0.389	0.401	0.395	B2	5/8	3/8

“Test without 2 events” leaves out the real NE|HVAC run on 3 Oct and the 18 implausible NW|Emerg. Lighting hours. It is a sensitivity view only; the official results keep them.

Interpretation

- **+1h:** clear gain over the Task 7 baselines (better on 7 of 8 circuits and on 8 of 8 days in both weeks). Against persistence: better on validation (0.246 vs 0.290); on the official test week persistence is ahead (0.930 vs 1.032), without the two events ML is ahead (0.265 vs 0.336).
- **+6h and day-ahead:** no measurable gain yet — level with the weekday × hour median (0.398 vs 0.390 and 0.395 vs 0.389 without the events).
- **Why:** with 35 days of training each weekday appears 5 times, so the model mostly learns “tomorrow looks like today at this circuit's level” (same hour yesterday carries 55–62% of input importance at +6h and day-ahead). It cannot foresee changes in operation such as NW|HVAC switching to standby and back.
- **Two events dominate the official test score:** NE|HVAC and NW|Emerg. Lighting produce 69–77% of the ML model's scaled test error.

Risks of the training-only rule

- The test week starts 8 days after the last training day; the model never learns from the validation week (recent values still reach it as inputs).
- The standby pattern of NW|HVAC in validation is never learned; only 35 days (about 710 hours per circuit) are available.
- Benefit: baselines are fitted the same way, so the comparison stays like-for-like.

4 · Open questions and Sprint 3

Items marked *(review)* come from the review of the Task 8 results and are not yet written into README_Task8.md.

Open point	Where it stands
Data settings (gap and coverage limits)	Proposed values in config.py; need team sign-off (README_Task7.md).
SE / SW circuits	Excluded; energy and power readings disagree; question for the data owner.
NW|Emerg. Lighting 5–6 Oct	Implausible readings kept in results; add a rule-based data-quality flag in dataset v1.2.
Protocol §9 text	Reworded rule agreed 18 Oct; Task 6 document still needs the update and a change note.
ML vs persistence at +1h (review)	Mixed on the official test week; state it without relying on the event-free view.
Under-forecasting (review)	MAE loss forecasts the typical hour: total day-ahead energy about 7% (validation) and 10% (test) below actual; peaks under-forecast.
Selection metric (review)	The hand dryer makes up 31–45% of the score used to choose settings; use a median or consumption weights.
Settings at the grid edge (review)	Simplest settings won everywhere; extend the grid towards more regularisation.
Test week already inspected (review)	Events were found during Task 7; keep 6–13 Oct as a fresh test week for Sprint 3.

Sprint 3 improvement list (from README_Task8.md)

- 1. More history: use the weeks after 5 Oct with rolling evaluation and periodic refits (protocol amendment needed); the 2020–2023 hourly history (about 25,000 hours per circuit) is the largest untapped source.
- 2. Day-ahead model that builds on simple rules (predict the change from the last value, or the deviation from the weekday × hour profile).
- 3. Operating-state and context inputs: on/off detection for HVAC, academic calendar, outdoor weather forecasts.
- 4. Data-quality rule for energy far above power; investigate SE/SW.
- 5. Pre-trained comparison (Chronos-Bolt, no training); 6. forecast ranges for anomaly detection; 7. W as second target.

5 · The ZIP package

36 files, 2.7 MB. “Minimum” marks what Tasks 7 and 8 need for their Definitions of Done; “recommended” is the evidence for no leakage; “optional” can be regenerated by the scripts.

```
Task8_first_ML_model.zip
|-- requirements.txt, .gitignore          setup
|-- config.py                             settings + team decisions
|-- task7_pipeline.py, task7_baselines.py, task8_model.py     building blocks
|-- run_task7.py ... run_task8_analysis.py                    the 5 scripts, run in order
|-- self_check.py                         hand-checks for VSCode
|-- README_Task7.md, README_Task7_baselines.md, README_Task8.md
|-- tests/      3 test files, 26 tests
|-- derived/    hourly dataset v1.1 + META + eligibility
|-- models/     frozen config + 3 models
|-- results/    predictions, scores, comparison tables
(data/ is not included: put Storey1_ALL_ZONES_FROZEN_v1.0.csv.xz or .csv.gz + manifest there)
```

File	Task	What it is	Size	Minimum
requirements.txt	Setup	Exact library versions the model was frozen with	0.2 KB	yes
.gitignore	Setup	Keeps .venv and cache folders out of the repository	0.1 KB	optional
config.py	7 + 8	All settings and team decisions in one place	4.5 KB	yes
task7_pipeline.py	7	Builds the hourly dataset (S05–S11)	14.8 KB	yes
task7_baselines.py	7	Forecast timing, Baselines 1–2, scoring, results table	10.6 KB	yes
task8_model.py	8	Model inputs, persistence baseline, gradient-boosting model	9.2 KB	yes
run_task7.py	7	Runs the dataset build from the frozen file	2.1 KB	yes
run_task7_baselines.py	7	Runs both baselines on validation and test	4.1 KB	yes
run_task8_tune.py	8	Trains, chooses on validation, freezes; refuses to re-tune	5.1 KB	yes
run_task8_test.py	8	Runs the frozen models once on the test week; checks checksums	4.7 KB	yes
run_task8_analysis.py	8	Error analysis behind the README tables	6.0 KB	yes
self_check.py	7 + 8	29 hand-checks (35 with a rebuild), run cell by cell in VSCode	8.0 KB	recommended
tests/test_task7.py	7	14 tests: hourly sums, missing never 0, outage rules, splits, metrics	5.7 KB	recommended
tests/test_task7_baselines.py	7	7 tests: forecast timing, baselines, training-only fit, scoring	4.1 KB	recommended
tests/test_task8.py	8	5 tests: no future data in inputs, training-only scale, determinism	3.5 KB	recommended
README_Task7.md	7	Hourly dataset v1.1: steps, findings, open settings	3.9 KB	yes
README_Task7_baselines.md	7	Task 7 deliverable: decisions, results, interpretation	7.8 KB	yes
README_Task8.md	8	Task 8 deliverable: setup, frozen model, comparison, Sprint 3	12.5 KB	yes
derived/Storey1_hourly_v1.1.parquet	7	Hourly dataset (rebuildable byte-identically in about 25 s)	413.0 KB	optional
derived/Storey1_hourly_v1.1_META.json	7	Checksums and settings of the dataset	2.9 KB	yes
derived/S10_eligibility_v1.1_THRESHOLD30.xlsx	7	Eligible / excluded circuits and threshold sensitivity	8.1 KB	yes
models/task8_frozen_config.json	8	Chosen settings, checksums, versions, training cut-offs	4.6 KB	yes
models/task8_p1h.joblib	8	Frozen model +1h	586.5 KB	yes
models/task8_p6h.joblib	8	Frozen model +6h	580.6 KB	yes
models/task8_day-ahead.joblib	8	Frozen model day-ahead	572.2 KB	yes
results/task7_results.csv	7	Baseline MAE/RMSE per circuit, horizon, week	15.7 KB	yes
results/task7_predictions_v1.1.parquet	7	Every baseline forecast	37.6 KB	yes

File	Task	What it is	Size	Minimum
results/task7_comparison.xlsx	7	Baseline comparison for reading in Excel	20.6 KB	optional
results/task8_results.csv	8	All four models, validation and test, same hours	60.5 KB	yes
results/task8_test_predictions.parquet	8	Every test-week forecast of all four models	78.1 KB	yes
results/task8_validation_predictions.parquet	8	Validation forecasts (input to the test script)	79.9 KB	yes
results/task8_validation_results.csv	8	Validation scores (input to the test script)	31.3 KB	yes
results/task8_tuning_log.csv	8	Validation score of all 48 configurations	9.8 KB	yes
results/task8_comparison.xlsx	8	Comparison, wins, coverage, frozen config in Excel	56.3 KB	optional
results/task8_error_analysis.xlsx	8	Overview, per circuit, by hours ahead, input importance	10.4 KB	optional
results/task8_tables.md	8	The tables quoted in README_Task8.md	5.3 KB	optional

Run and verify

```
pip install -r requirements.txt
python -m pytest -q tests/                  # 26 passed
python run_task7.py --data data/Storey1_ALL_ZONES_FROZEN_v1.0.csv.xz --out derived
python run_task7_baselines.py
python run_task8_tune.py                    # in a fresh folder; refuses once frozen
python run_task8_test.py
python run_task8_analysis.py
python self_check.py [--fresh <rebuilt folder>]   # 29 of 29 (35 of 35)
```

Step-by-step instructions with expected outputs: the separate self-check guide for VSCode.